D 103547 **(Pages : 4)** **Name.....**

Reg. No.....

SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2024

Mathematics

MAT 2B 02—CALCULUS

(2014—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type Questions)

Answer **all** *questions.*
Each question carries 1 *mark.*

1. What is the absolute maximum of $f(x) = x^2$ on the domain $D = (0, 2]$.

2. Evaluate $\lim_{x \to -\infty} \frac{\pi\sqrt{3}}{x^2}$.

3. Evaluate $\sum_{k=1}^{4} \cos k\pi$.

4. Suppose that f and h are integrable and that $\int_7^9 f(x)\,dx = 5$, $\int_7^9 h(x)\,dx = 4$. Evaluate $\int_9^7 [h(x) - f(x)]\,dx$.

5. Find the average value of the function $f(t) = (t-1)^2$ on $[0, 3]$.

6. Use the Fundamental Theorem to find dy/dx if $y = \int_x^5 3t \sin t\,dt$.

7. How do you define and calculate the area of the region between the graphs of two continuous functions ?

Turn over

8. How do you define and calculate the length of the graph of a smooth function over a closed interval ?

9. How do you define and calculate the area of the surface swept out by revolving the graph of a smooth function $y = f(x), a \le x \le b,$ about the x-axis ?

10. What is the Center of mass of a thin road along the x-axis with density function $\delta(x)$?

11. Define the work done by a variable force $F(x)$ directed along the x-axis from $x = a$ to $x = b$.

12. State Hooke's Law for springs.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** *questions.*
Each question carries 2 *marks.*

13. State Rolle's Theorem.

14. Verify Mean Value Theorem for the function $f(x) = \sqrt{x-1},$ in the interval $[1, 3]$.

15. Find the linearization of $f(x) = x^3 - 2x + 3$, at $x = 2$.

16. Evaluate $\int_{-\pi/3}^{-\pi/4} \left(4 \sec^2 t + \frac{\pi}{t^2}\right) dt.$

17. Using substitution evaluate the integral $\int_0^{\pi} 3 \cos^2 x \sin x dx.$

18. Find the area of the region enclosed by the line $y = 8x$ and curve $y = x^4$.

19. Find the volume of the solid generated by revolving the region between the y-axis and the curve $x = 2/y, 1 \le y \le 4,$ about the y-axis.

20. Set up an integral for the length of the curve $x = \sin y$ in the interval $0 \le y \le \pi$.

21. Set up an integral for the area of the surface generated by revolving the curve $y = x^2,\ 0 \le x \le 2$ about x-axis.

22. Show that the center of mass of a straight, thin strip or rod of constant density lies halfway between its two ends.

23. A mountain climber is about to haul up a 50 m length of hanging rope. How much work will it take if the rope weighs 0.624 N/m ?

24. What is the Center of Mass of a thin plate covering a region in the xy-plane ?

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Given $f'(x) = (x-1)(x+2)$.

    (a) What are the critical points of f?

    (b) On what intervals is f increasing or decreasing ?

26. Using the Sandwich Theorem, find the asymptotes of the curve

$$y = 2 + \frac{\sin x}{x}.$$

27. State and prove the First Derivative Theorem for Local Extreme Values.

28. The revenue and cost functions at a Gadget company are

$r(x) = 9x$ and $c(x) = x^3 - 6x^2 + 15x$,

where x represents thousands of gadgets. Is there a production level that will maximize the company's profit ? If so, what is it ?

29. Find the total area between the region $y = 3x^2 - 3, \quad -2 \le x \le 2$ and the x-axis.

30. Find the volume of the solid lies between planes perpendicular to the x-axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x-axis between these planes are squares whose bases run from the semicircle $y = -\sqrt{1 - x^2}$ to the semicircle $y = \sqrt{1 - x^2}$.

31. Find the length of the curve $y = x^{3/2}$ from $x = 0$ to $x = 4$.

Turn over

32. Find the volume of the solid generated by revolving the region between the y-axis and the curve $x = 2/y, 1 \le y \le 4$, about the y-axis.

33. Find the center of mass of a wire of constant density δ shaped like a semicircle of radius a.

(6 × 5 = 30 marks)

Part D (Essay Type)

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. (a) Find the intervals on which $g(t) = -3t^2 + 9t + 5$ is increasing and decreasing. Where does the function assume extreme values and what are these values ? (5 marks)

(b) Sketch the curve $y = x^2 - 4x + 3$. Include the co-ordinates of any local and absolute extreme points and inflection points in your sketch. (5 marks)

35. (a) If f is continuous at every point of $[a, b]$ and F is any antiderivative of f on $[a, b]$, then prove that

$$\int_a^b f(x)\,dx = \mathrm{F}(b) - \mathrm{F}(a).$$ (5 marks)

(b) The edge x of a cube is measured with an error of at most 0.5 %.

i) What is the maximum corresponding percentage error in computing the cube's surface area ?

ii) What is the maximum corresponding percentage error in computing the cube's volume ?

(5 marks)

36. (a) Find the area of the surface generated by revolving the curve $y = x^3, 0 \le x \le 1/2$, about the x-axis. (5 marks)

(b) A force of 2 N will stretch a rubber band 2 cm (0.02 m). Assuming that Hooke's Law applies, how far will a 4-N force stretch the rubber band ? How much work does it take to stretch the rubber band this far ? (5 marks)

[2 × 10 = 20 marks]